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BSCS – 3B

Task 1:

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import geopandas as gpd

print("✅ Libraries imported successfully.\n")
```

```
✅ Libraries imported successfully.
```

Task 2: Load Natural Earth Dataset

1. How many columns does the dataset contain?
   169 columns (based on your printed output).
2. What type of geometries are included?
   Polygon and MultiPolygon (countries can be single-shape or multiple islands/parts).
3. What does the geometry column represent?
   The geometry column stores the actual spatial shape/boundary of each country (its coordinates forming polygons).

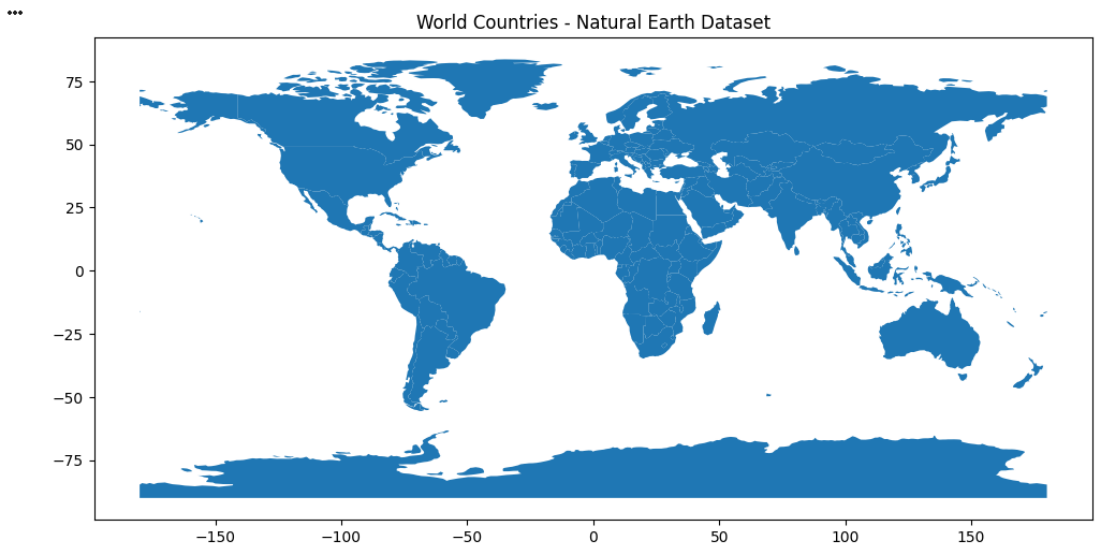
```python
url = "https://naturalearth.s3.amazonaws.com/110m_cultural/ne_110m_admin_0_countries.zip"
world = gpd.read_file(url)

print("Number of columns:", len(world.columns))
print("Geometry types:", world.geom_type.unique())

world.head()
```

```
of columns: 169
ry types: ['MultiPolygon' 'Polygon']
```

aturecla	scalerank	LABELRANK	SOVEREIGNT	SOV_A3	ADM0_DIF	LEVEL	TYPE	TLC	ADMIN	...	FCLASS_TR	FCLASS_ID	FCLASS_PL	FCLASS_GR	FCLASS_IT	FCLASS_NL	FCLASS_SE	FCLASS_BD	FCLASS_UA	geometry
Admin-0 country	1	6	Fiji	FJI	0	2	Sovereign country	1	Fiji	...	None	None	None	None	None	None	None	None	None	MULTIPOLYGON (((180 -16.06713, 180 -16.55522, ...
Admin-0 country	1	3	United Republic of Tanzania	TZA	0	2	Sovereign country	1	United Republic of Tanzania	...	None	None	None	None	None	None	None	None	None	POLYGON ((33.90371 -0.95, 34.07262 -1.05982, 3...
Admin-0 country	1	7	Western Sahara	SAH	0	2	Indeterminate	1	Western Sahara	...	Unrecognized	Unrecognized	Unrecognized	None	None	Unrecognized	None	None	None	POLYGON ((-8.66559 27.65643, -8.66512 27.58948...
Admin-0 country	1	2	Canada	CAN	0	2	Sovereign country	1	Canada	...	None	None	None	None	None	None	None	None	None	MULTIPOLYGON (((-122.84 49, -122.97421 49.0025...
Admin-0 country	1	2	United States of America	US1	1	2	Country	1	United States of America	...	None	None	None	None	None	None	None	None	None	MULTIPOLYGON (((-122.84 49, -120 49, -117.0312...

× 169 columns

```
world.plot(figsize=(12,6))
plt.title("World Countries - Natural Earth Dataset")
plt.show()
```



Task 3: Check the Coordinate Reference System (CRS)

1. **What does EPSG:4326 represent?**
   WGS84 Geographic CRS using latitude and longitude coordinates in degrees.
2. **Why is CRS important in spatial analysis?**
   Because CRS controls how locations are measured and displayed—wrong CRS can cause incorrect distance/area results.

```
print("Current CRS:", world.crs)
```

```
Current CRS: EPSG:4326
```

Task 4: Convert to Metric CRS for Area Calculation

1. **Why can't we compute area accurately using EPSG:4326?**
   Because EPSG:4326 uses degrees, not meters, and degrees are not a consistent unit of distance/area, so area results become inaccurate.
2. **What unit is EPSG:3857 based on?**
   Meters (m), so computed area becomes square meters (m^2).

```python
world_projected = world.to_crs(epsg=3857)

world_projected["area_m2"] = world_projected.geometry.area

world_projected[["ADMIN", "area_m2"]].head()
```

	ADMIN	area_m2
0	Fiji	2.128334e+10
1	United Republic of Tanzania	9.522552e+11
2	Western Sahara	1.171023e+11
3	Canada	5.216648e+13
4	United States of America	2.186228e+13

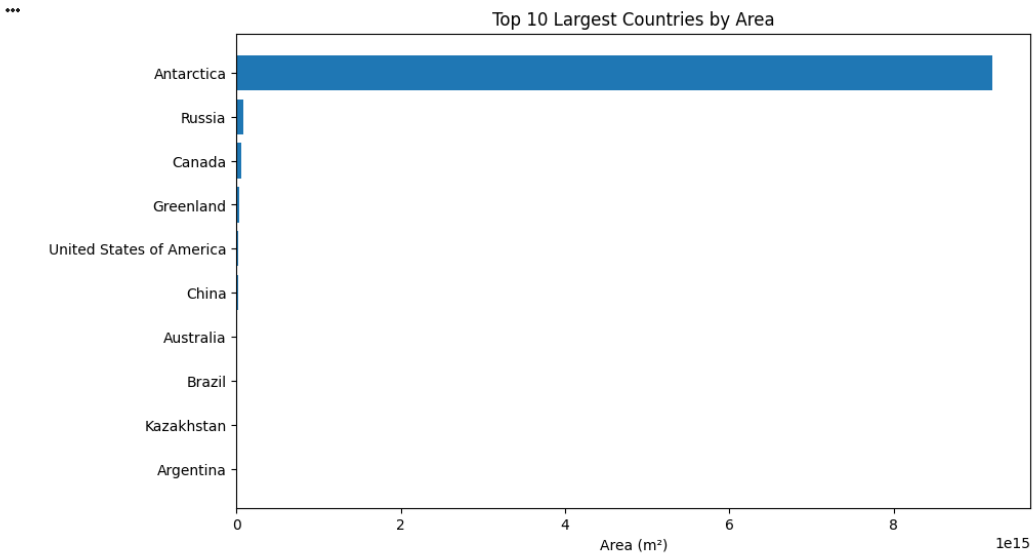
Task 5: Extract Centroid Coordinates

1. **What is a centroid?**
   A center point of a geometry (like the “balance point” of a country’s shape).
2. **How can centroid coordinates be used in clustering?**
   They can represent each country as a single point (x, y) so algorithms like clustering can group countries based on spatial location.
3. **Why do we extract X and Y separately?**
   Because most analysis/ML methods require separate numeric features, so x and y are used as two input variables.

```python
world_projected["centroid"] = world_projected.geometry.centroid
world_projected["x"] = world_projected.centroid.x
world_projected["y"] = world_projected.centroid.y

world_projected[["ADMIN", "x", "y"]].head()
```

	ADMIN	x	y
0	Fiji	1.824878e+07	-1.958098e+06
1	United Republic of Tanzania	3.869296e+06	-7.003071e+05
2	Western Sahara	-1.348403e+06	2.794163e+06
3	Canada	-1.079779e+07	1.044422e+07
4	United States of America	-1.329713e+07	6.667416e+06

```python
top10 = world_projected.sort_values(by="area_m2", ascending=False).head(10)

plt.figure(figsize=(10,6))
plt.barh(top10["ADMIN"], top10["area_m2"])
plt.xlabel("Area (m²)")
plt.title("Top 10 Largest Countries by Area")
plt.gca().invert_yaxis()
plt.show()
```



Reflection (3–5)

During this lab, I had some difficulty understanding why area calculations were wrong at first, but I learned that CRS affects measurement accuracy. I realized that EPSG:4326 uses degrees, so it is not suitable for computing real-world areas. Converting the dataset to a metric CRS like EPSG:3857 helped me compute areas in square meters correctly. Overall, I learned that spatial data needs proper projection before doing analysis like area computation and centroid extraction.